

CSE 340: Computer Architecture

[Spring 2025]

Quiz - 4 :: Set - A

Time: 15 min

Marks: 15

Name: _____

ID: _____ Section: _____

1. Assume pipelined execution with the classic 5-stage RISC-V pipeline (IF, ID, EX, MEM, WB). Consider the following sequence of instructions:

ADD x19, x20, x21

LD x21, 0(x22)

SUB x20, x19, x21

LD x23, 4(x20)

LD x22, 0(x23)

Assume that data forwarding is available whenever possible, but stalls must be introduced if a hazard cannot be resolved through forwarding.

Tasks:

- a. Draw the multi-cycle pipeline execution diagram for the given sequence of instructions.
- a. Clearly indicate all stalls and label the forwarding paths used to resolve data hazards.
- b. Briefly explain why stalls are introduced and how data forwarding reduces the number of stalls required.

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Quiz - 4 :: Set - B

Time: 15 min

Marks: 15

Name: _____

ID: _____ Section: _____

1. Assume pipelined execution with the classic 5-stage RISC-V pipeline (IF, ID, EX, MEM, WB). Consider the following sequence of instructions:

XOR x19, x22, x21

LD x21, 0(x22)

OR x20, x19, x21

LD x23, 4(x21)

LD x22, 0(x20)

Assume that data forwarding is available whenever possible, but stalls must be introduced if a hazard cannot be resolved through forwarding.

Tasks:

- b. Draw the multi-cycle pipeline execution diagram for the given sequence of instructions.
- c. Clearly indicate all stalls and label the forwarding paths used to resolve data hazards.
- d. Briefly explain why stalls are introduced and how data forwarding reduces the number of stalls required.